VFX Tool Kit

InnoGames

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Introduction



Introduction

The project ‘VFX HeroCity Playground’ includes the main scene with in-game lighting and backgrounds. Within this project, you’ll find shaders, scripts, and import rules for geometry and textures.

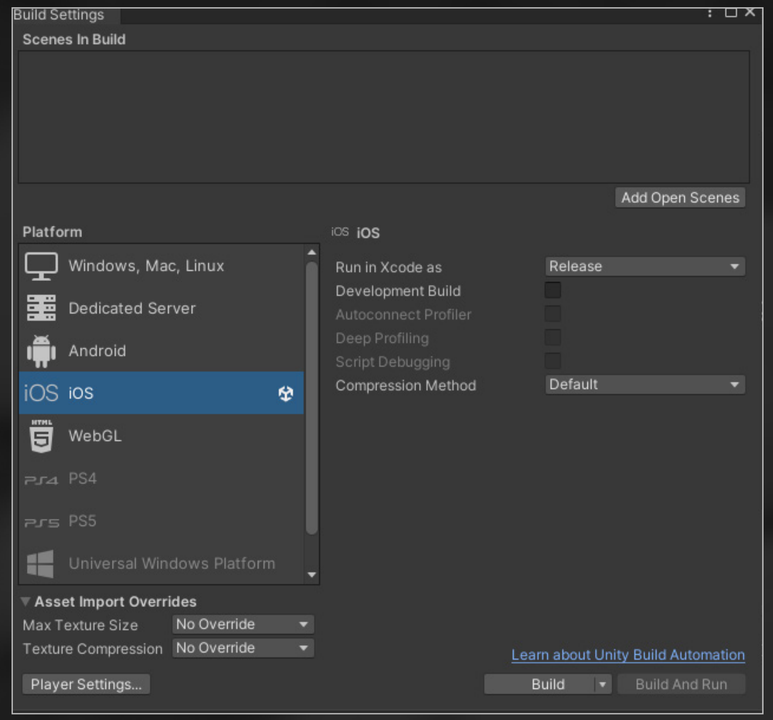
A complete ability scene VFX_King_Minos.

For more specific examples regarding scripts and shaders, refer to:

- **‘VFX_Script_Shader_Example_Scene’**
- **‘VFX_Shader_Properties’**
-

Additionally, there’s a pilot scene named ‘VFX_Achilles.’”

Project Installation



Install

Download Unity 2022.3.6f from here. https://unity.com/releases/editor/archive

Unzip the Assets File 'VFX_InnoGames_ToolKit' in to your Unity project/Assets.

NOTE: Set Build settings to iOS or Android.
File > Build Settings

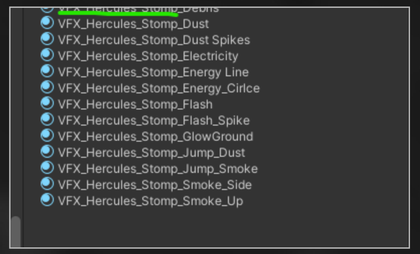
Naming Convetions



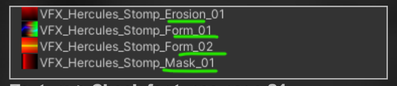
Hero Name > Take the name from the character folder.



Ability Type > You can find it here in the concept.



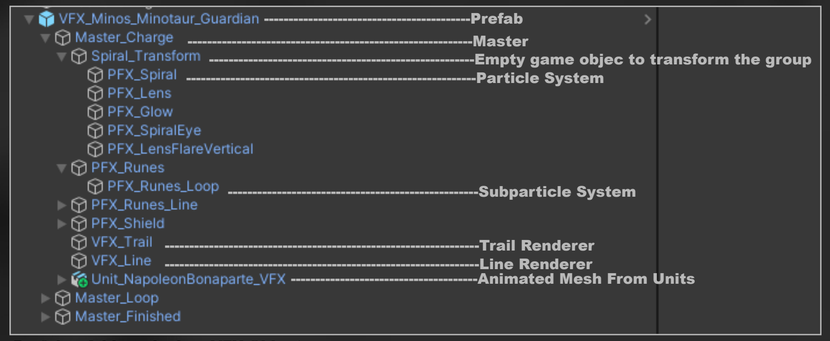
Material



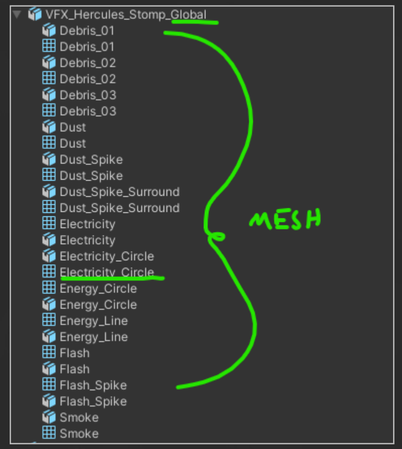
Texture > Check for types page 21.



FBX Types
Global > Page 18
Shape > Page 19
Volume > Page 12
VtxMotion > Page 14



Prefab > A hierachy in a VFX Object.



Mesh > Name

We are reusing the abilites, for that reason the naming conventions are importan to make sure the dublication process is efficent.

Asset Naming

Folder = (Hero Name)

Prefab = VFX_(Hero Name)_(Ability Type)_(Name)

Material = VFX_(Hero Name)_(Ability Type)_(Name)

Texture = VFX_(Hero Name)_(Ability Type)_(Texture Type)

FBX = VFX_(Hero Name)_(Ability Name)_(FBX Type)

Mesh = Name

Prefab inside naming

Master Object = Master_ OR Master
By mulitple Master Objects = Master_(Name)

Particle System = PFX_(Name)
Trail And Line Renderes = VFX_(Name)

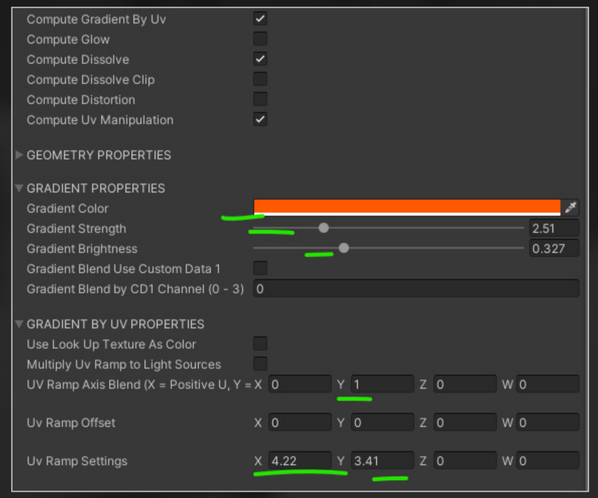
NOTE: Check the workflow tools on page 19 & 20.

Light And Gradients





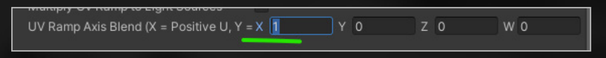
The gradient is using the first UV set of the mesh.



Uv Ramp Settings: X is Multiplier / Y is Power.



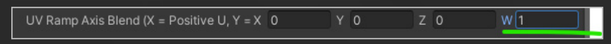
Setting the direction of the gradient by the UV Ramp Blend Properties.











Light And Gradients





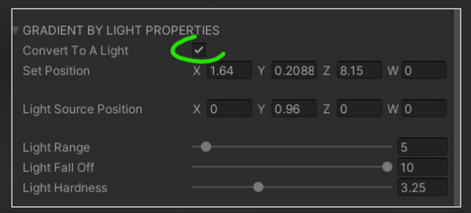


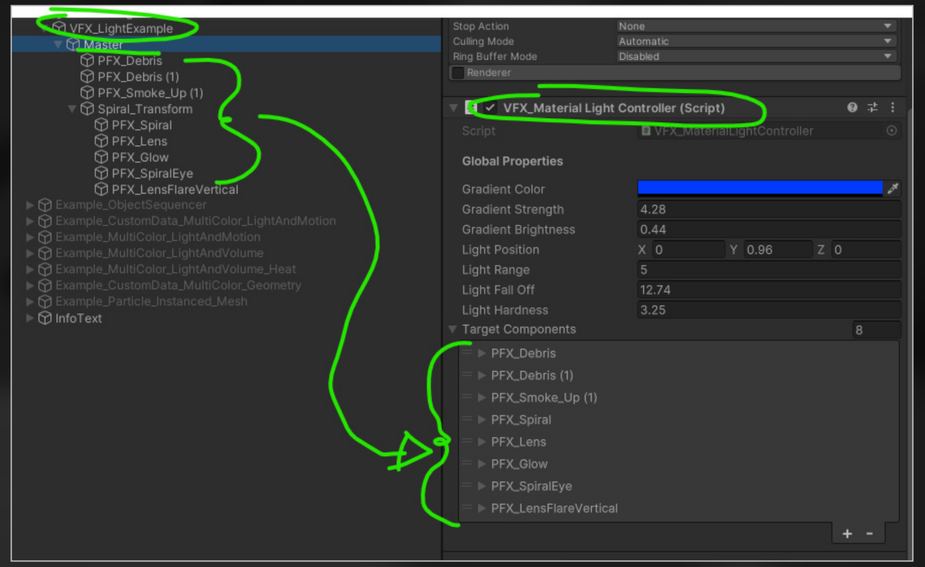
All particles in the Material Light Controller will have the same light source position.





By using ‘Converting To A Light’ the 3d Space gradient will behave like a light.



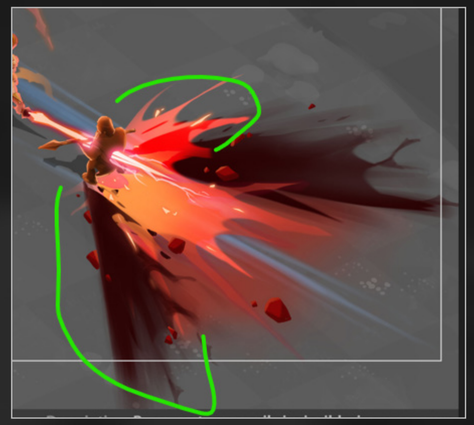


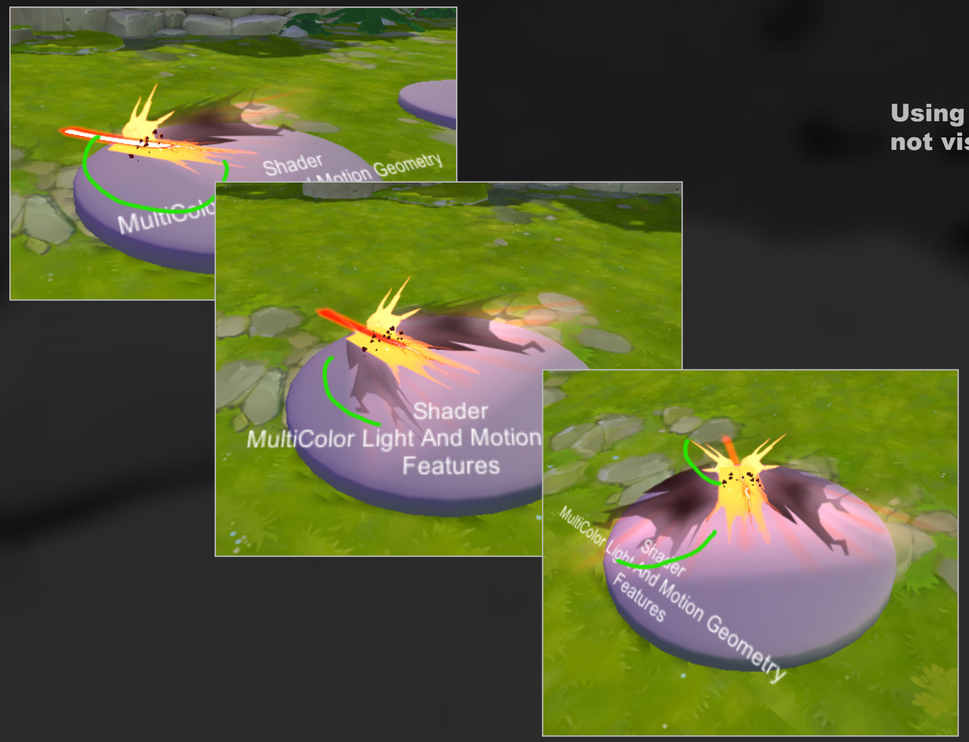
Add all particles to the Material Light Controller.

This script is setting the local position.

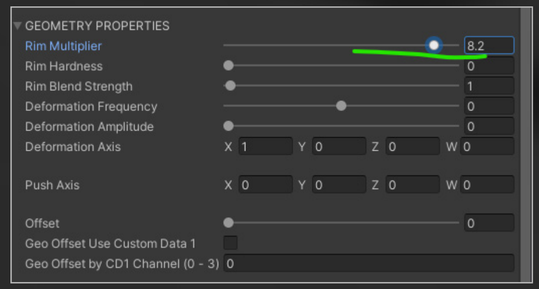
NOTE:
In the ‘Target Components’ you need enable ‘Use Global Settings’ to manipulate the light setting in the shader.

More Perspektive And Volume





Using geometry properties to make things visible or not visible relative to the normal and the camera direction.



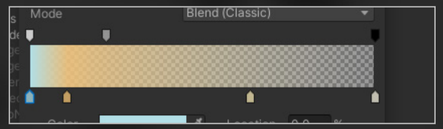
Glow And Premultiply



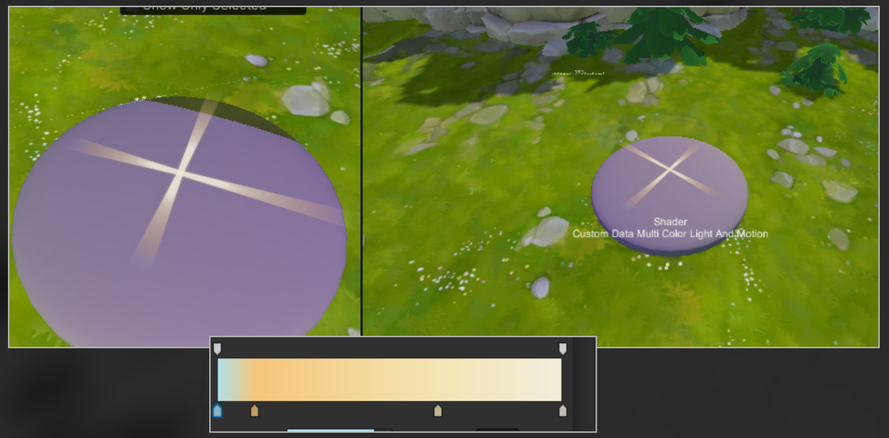
Just premultiplied effect.

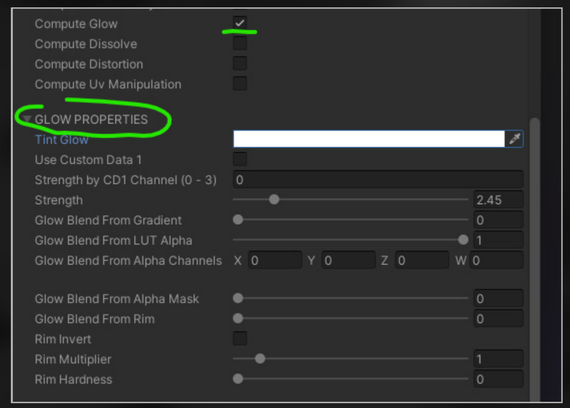
Premultiplied effect and glow.





0 Alpha is 100 % Premultiplied.
Is premultiplied active the values becomes higher as 1.0 so the glow is getting active.

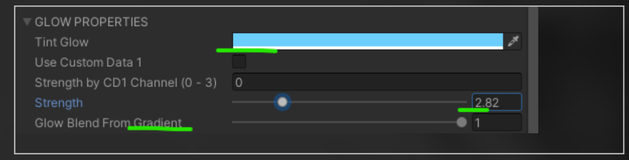








With the blending properties you can define where and how the glow should be visible.



Clouds / Smoke / Explosions

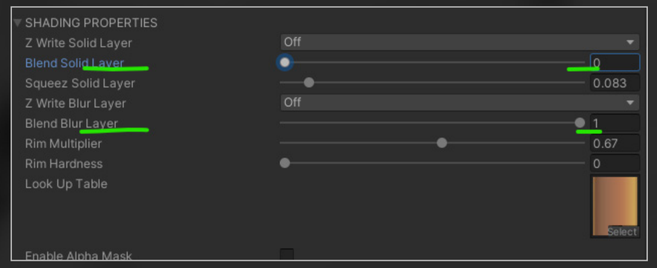




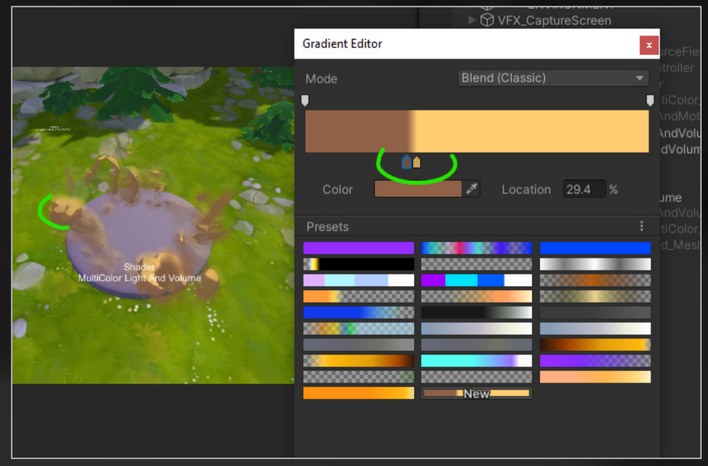


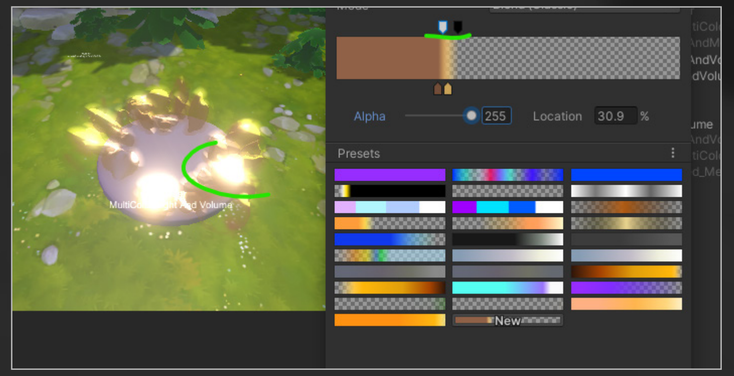






The blend layer properties helps to control how soft or how hard the smoke should be.





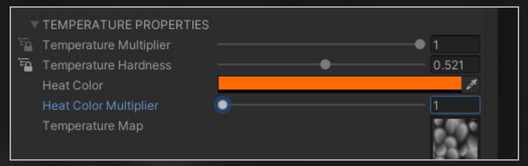
The ‘Look Up Texture’ is coloring the light from the main light in the scene.

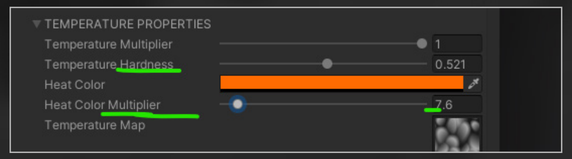
By changing the gradient you can control the sharpness of th light and the glowing areas.

Clouds / Smoke / Explosions









The ‘Heat Color Multiplier’ is setting the glow strength.

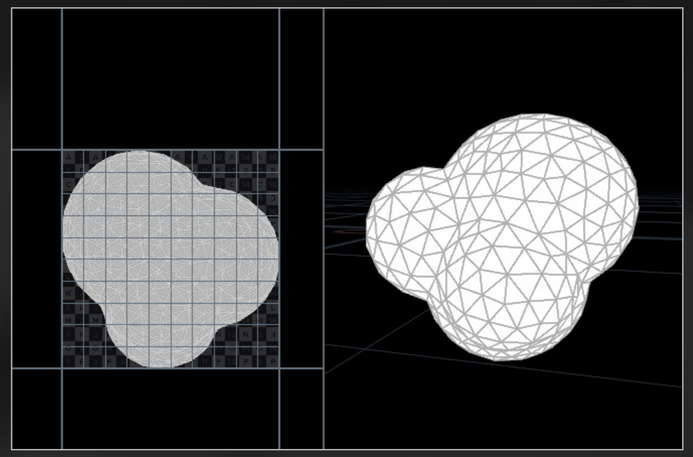
The temperature map is defining the dissolve and heat areas. Those options behave like the dissolve properties.







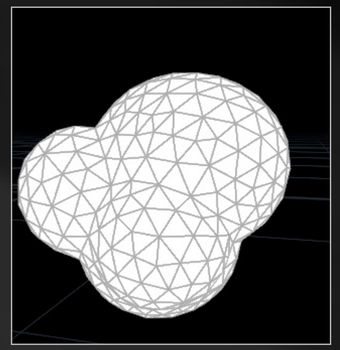
Clouds / Smoke / Explosions

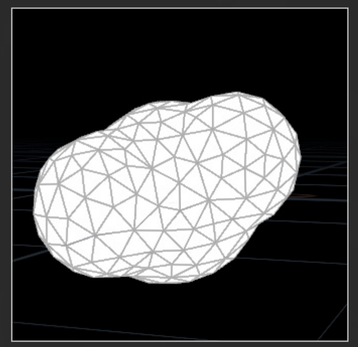


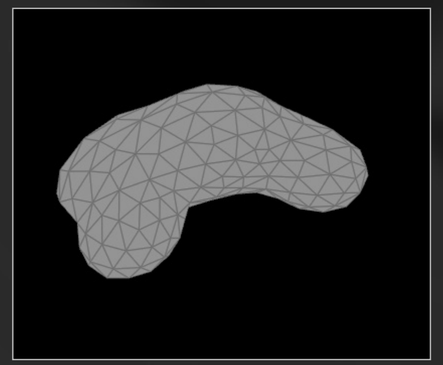
The uv should project from any side, just not from top or bottom.

The colud geometry should have unified topology.

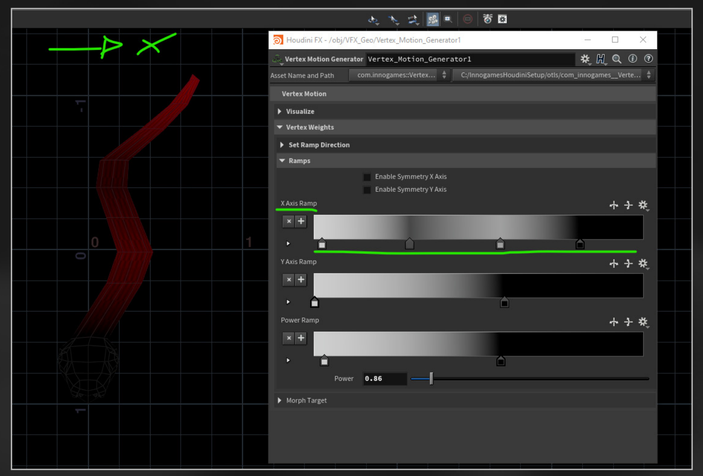
This important for the vertex calculations in the shader. Like for deformations and uv mappings.

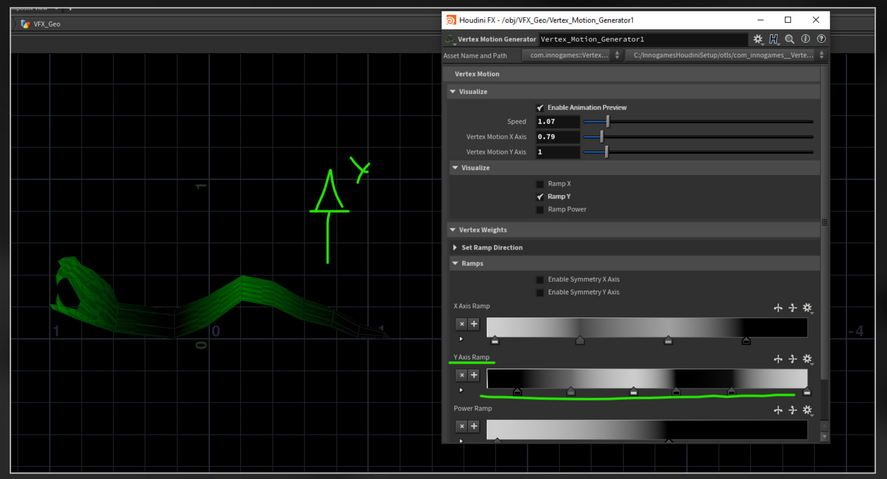


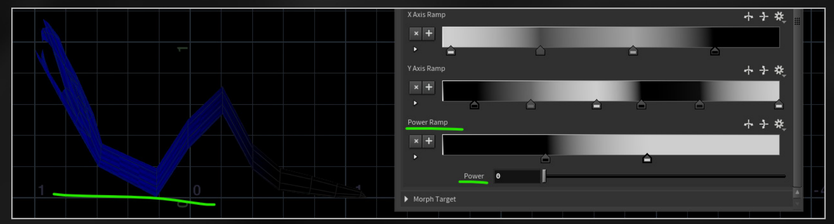


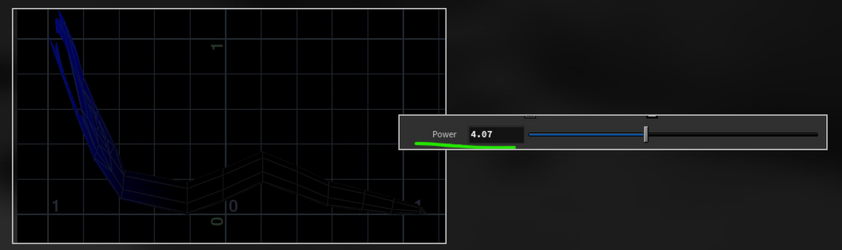


Simple Crowd Animations





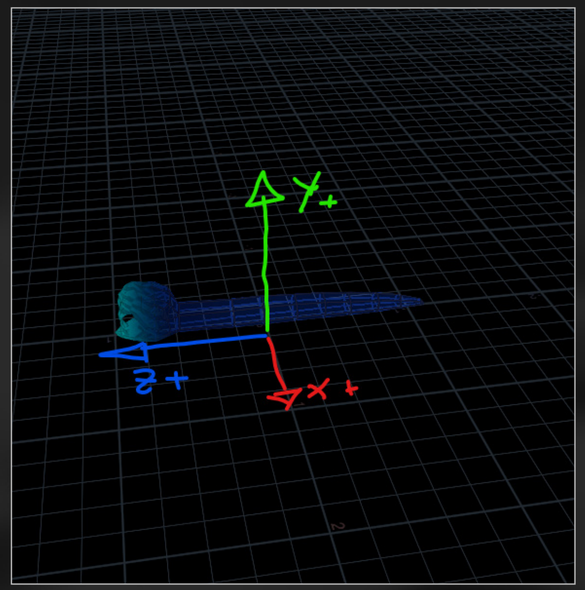




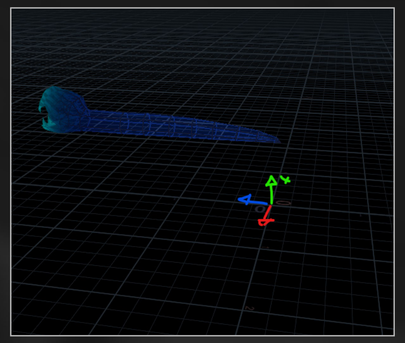
To get a simple ‘Sin Wave Animation’, the geometry needs vertex color weights.

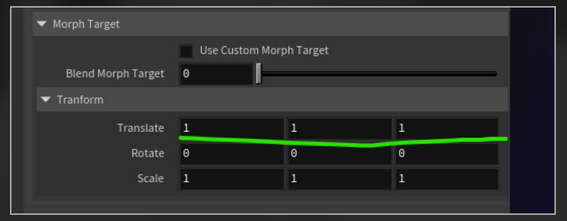
Red Channel = Weights for the X axis.
Green Channel = Weights for the Y axis.
Blue Channel = Weights for the Power.
Alpha Channel = The Power Value.

Simple Crowd Animations



Mesh pivot must be in the center.
Mesh look at direction should be posivtive Z.





The morph target is just a offset in all axis.

NOTE:
You can use a more complex morph target.

uv1[0]	uv1[1]	uv1[2]	uv2[0]	uv2[1]	uv2[2]
1.0	1.00012	0.0	-0.138835	0.0	0.0
1.05429	1.33816	0.0	1.90154	0.943229	0.0
1.0	0.990883	0.0	1.73969	0.474873	0.0
1.13369	1.00178	0.0	1.73942	0.493663	0.0
1.18514	1.09554	0.0	1.63951	0.292027	0.0
1.0	1.28186	0.0	1.61948	0.230037	0.0
1.13877	1.24787	0.0	1.66542	0.312864	0.0
1.0659	1.00579	0.0	1.01483	0.00130076	0.0
1.0	1.18875	0.0	1.015	3.52005e-05	0.0
1.0	0.995694	0.0	1.0148	0.00126103	0.0
1.11689	1.06587	0.0	1.01486	0.000836745	0.0
1.07802	1.14937	0.0	1.01488	0.000141745	0.0
1.02698	1.00579	0.0	-0.0931405	0.0	0.0
1.0	1.04558	0.0	-0.102829	0.0	0.0
1.0	0.998714	0.0	-0.0845871	0.0	0.0
1.03438	1.01812	0.0	-0.0973147	0.0	0.0
1.02772	1.03442	0.0	-0.100107	0.0	0.0
1.0	0.999271	0.0	0.235506	0.0	0.0
1.07987	1.04041	0.0	0.2252	0.0	0.0
1.06118	1.09891	0.0	0.218145	0.0	0.0
1.05796	1.00773	0.0	0.232709	0.0	0.0
1.0	1.12152	0.0	0.215624	0.0	0.0
1.0	0.996639	0.0	1.59173	0.21841	0.0

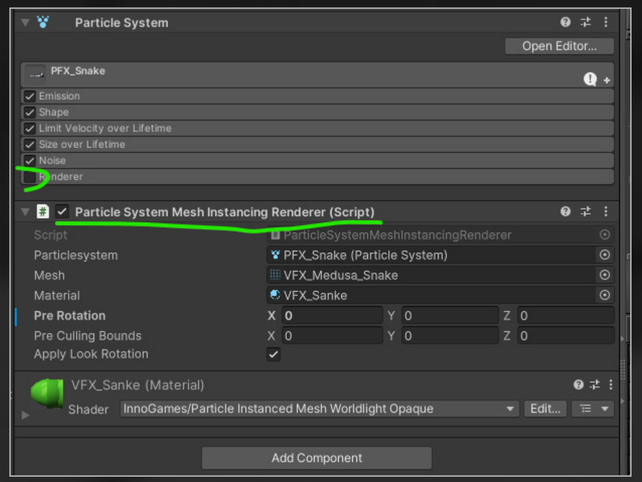
The morph target should stored in second and third UV set.

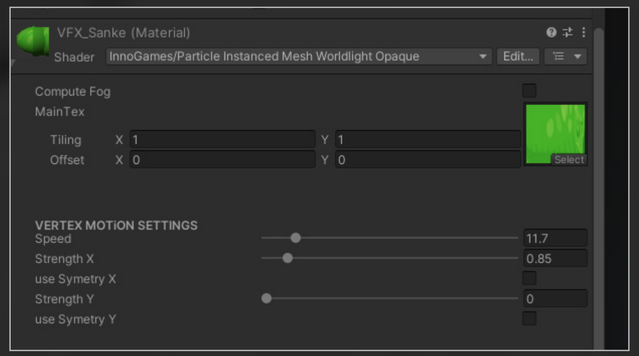
FBX is not capable to save four UV channels.

We need the UV Set 1 for X and Y,
and the UV Set 2 for Z.

Simple Crowd Animations



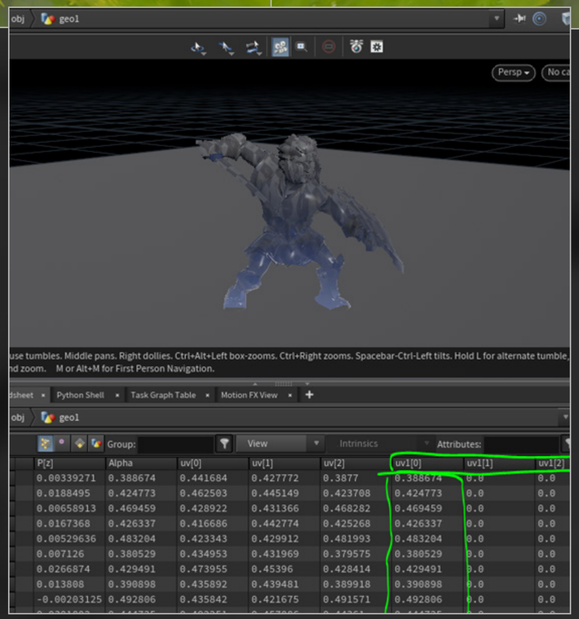




NOTE:
In the particle render module.
The render alignment should set on velocity.

Vertex Alpha





- Vertex Alpha

Particles are overwriting the vertex color and vertex alpha.
To sperate them, we are storing the vertex alpha in a second uv set in the index 0.

The shaders has a properties in the material render tab to enable and to disable the vertex alpha.

NOTE: If a particle billboard has a strange transparency gradient, this can be effected by the vertex alpha.
The vertex alpha should be disable by particle billboards.

GEOMETRY Vertex COLOR

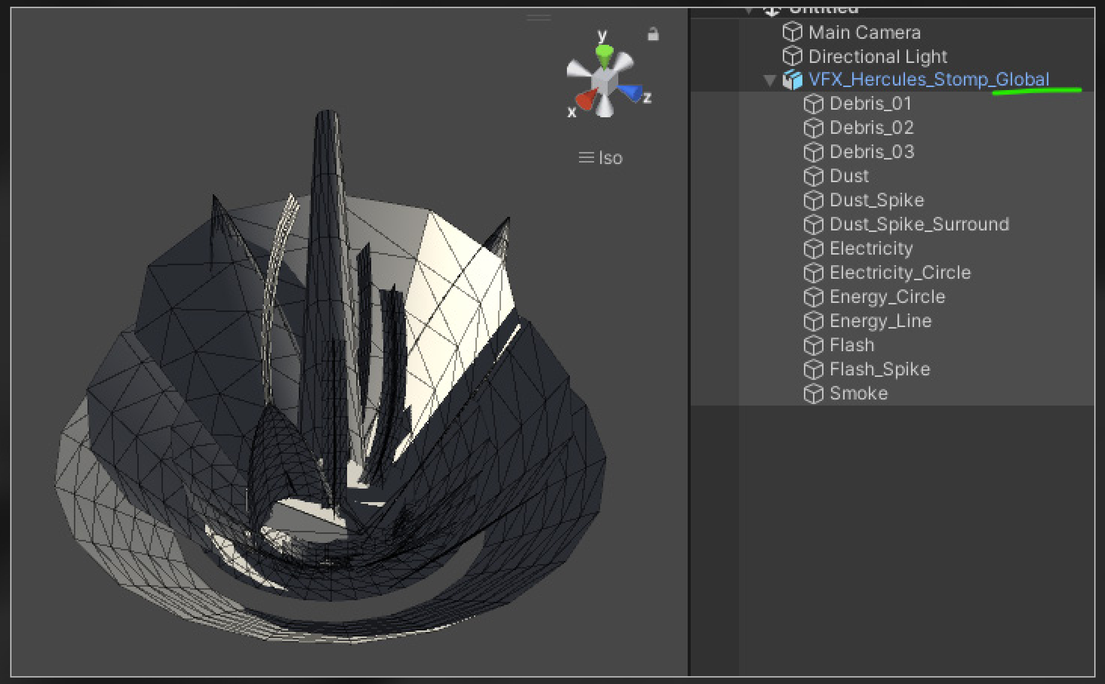


- **Vertex Color.**

  The shaders has a properties in the material shading tab to enable and to disable vertex color.

  The shader property ‘Multiply Particle Color’ will enable the vertex colors.

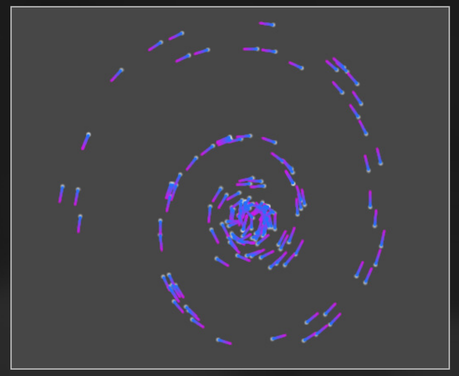
Global FBX

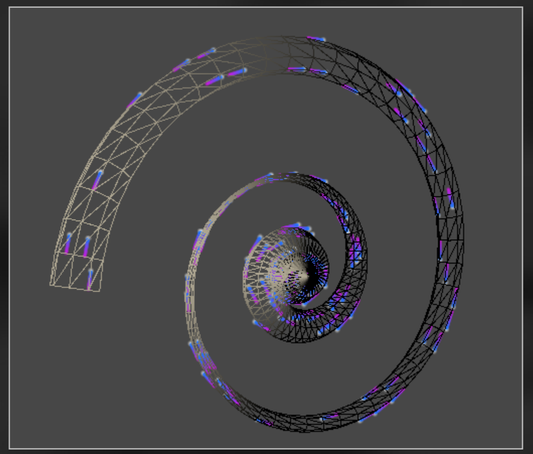


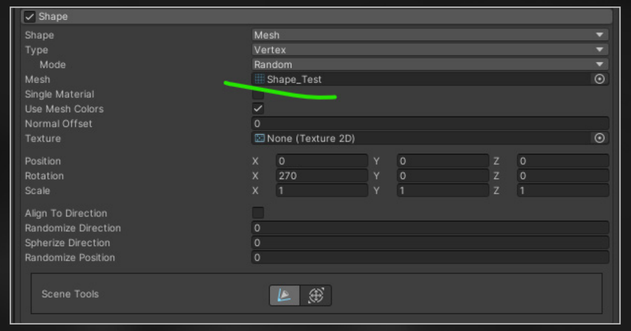
- Global FBX.

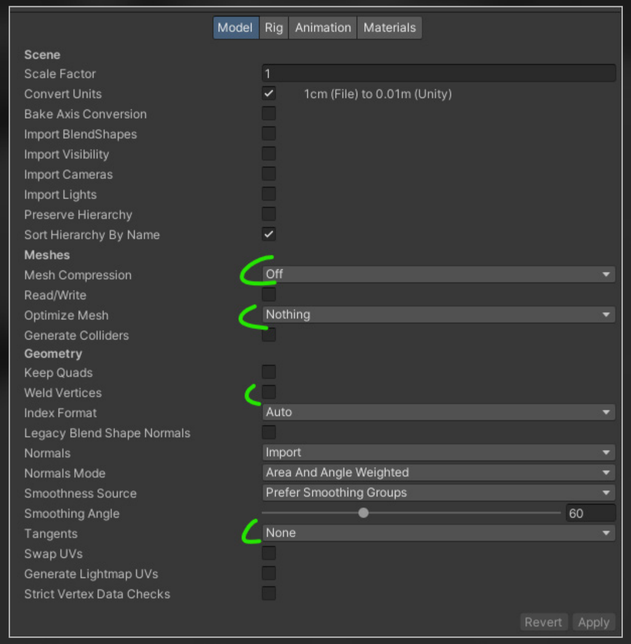
The global FBX includes all relevant meshes for a specific ability, and all meshes have a second UV set, which stores the vertex alpha in index [0].

Shape FBX









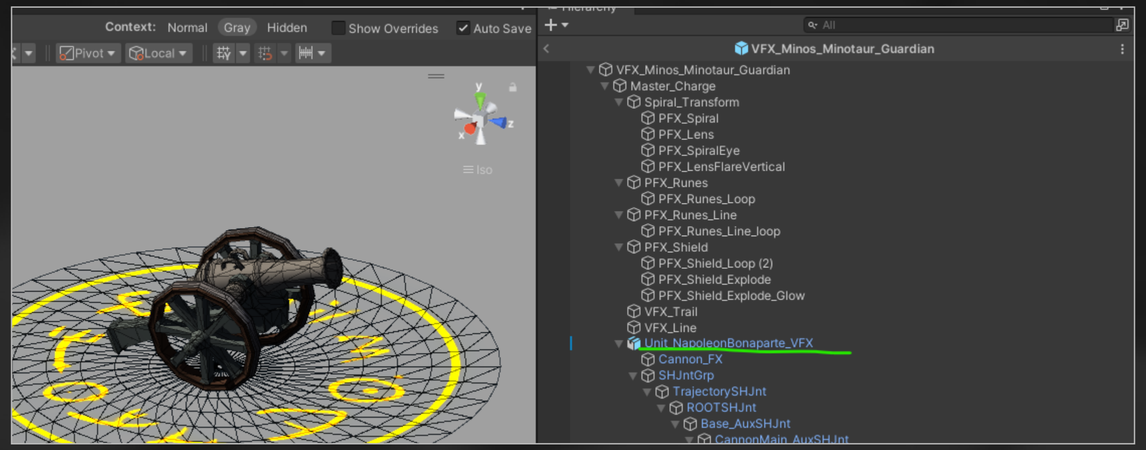
FBX import settings.

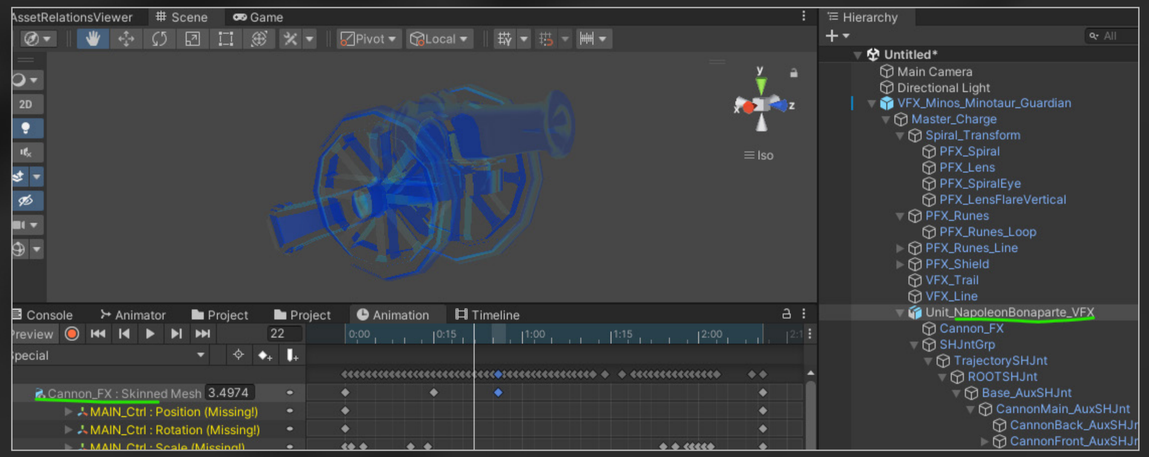
- Shape Mesh

Shape mesh is a mesh used to define the position and initial velocity direction.

NOTE: Shape meshes have specific FBX import settings to ensure that the import does not alter any vertex index and/or order.

GEOMETRY Animated Mesh

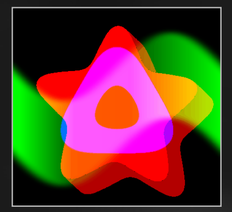




How to add?

- Animated meshes should be added under a master.
- Particles System / Trails / Line Renderer etc. can be added to bones.
- Material properties can be animated by keys.
- NOTE: Use ‘Multi Color Light And Motion’ shader to animated effects.

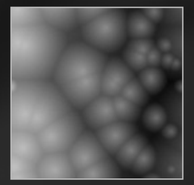
TEXTURE Types



- Packed Texture
usual size 128x128
max size 256x256
RGB: None compression RGB 24 bit



- Texture
usual size 128x128
max size 256x256
RGBA: None compression RGBA 32 bit.



- Dissolve, Ramp, Erosion Maps
usual size 128x128
max size 256x256
R: None compression R 8 bit.



- Distortion, Flow, Motion Vector Maps
usual size 64x64
max size 128x128
RGB: None compression RGB 24 bit.



- Mask Maps
usual size 64x64
max size 256x256
RGB: None compression RGB 24 bit.

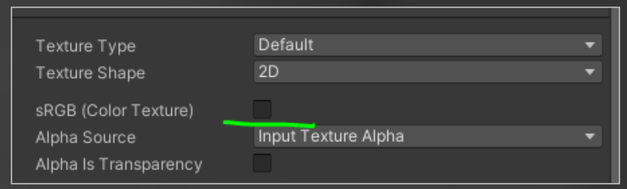
- **Packed Textures**

Are using the texture channels RGBA to store different type of textures. For preformance and disk space reasons we are using just RGB channels. The usual size is 128x128.

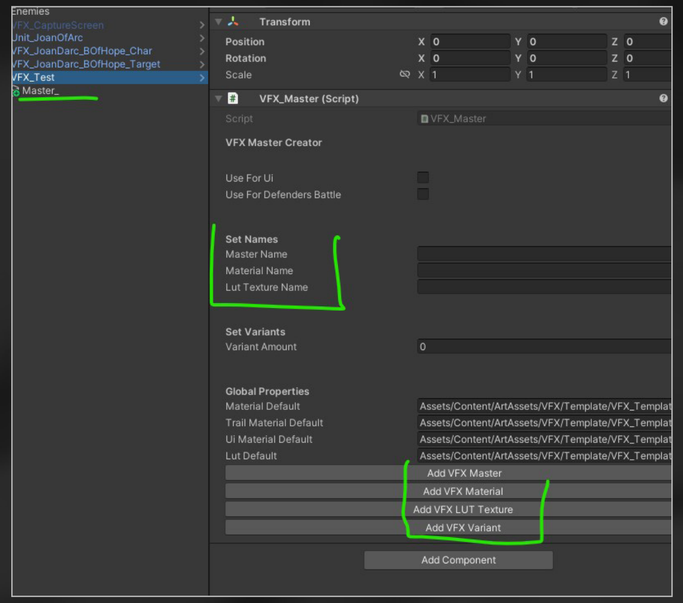
If is necessary to have more pixels for detailed, thin or to have sharper texture you can go with 256x256. Everything above is not recomanded with none compression.

RGB: None compression RGB 24 bit.

NOTE: sRBG should be disabled for distortion maps etc.



VFX Object Creation



Introduction

To make the workflow more streamlined and intuitive, we have VFX workflow scripts. These scripts create particle systems with specific modifications, such as those in the vertex stream settings.

They also assist in naming assets and adding assets to the VFX folder.

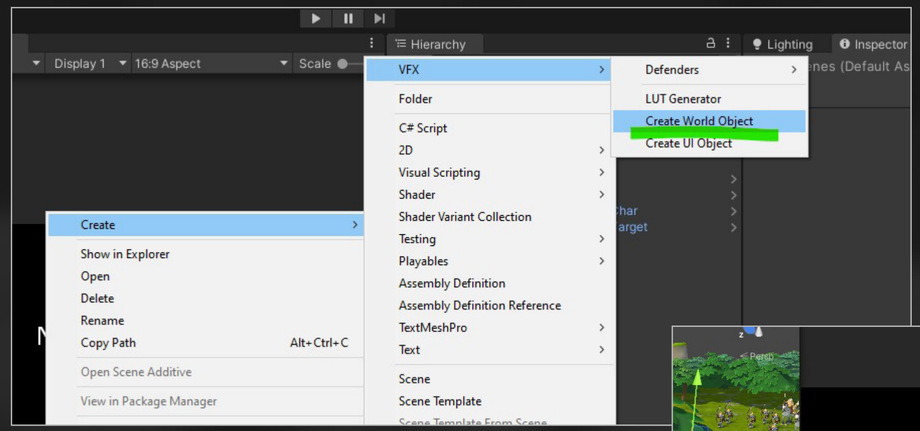
Under the hood, the script knows where the VFX object was created and generates a relative path for all other game assets, such as materials, lookup textures, etc.

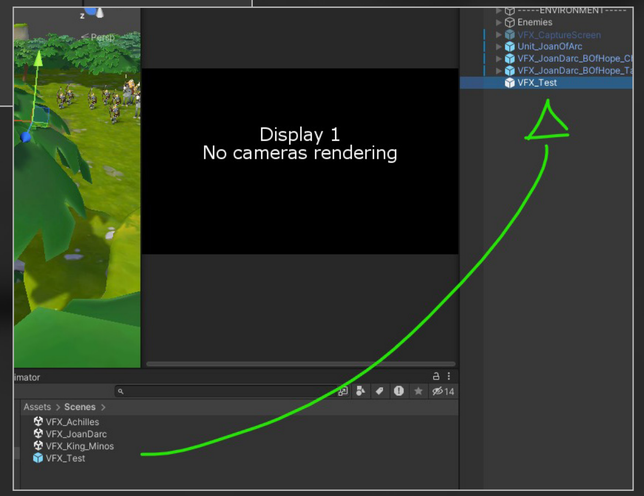
VFX Object Creation

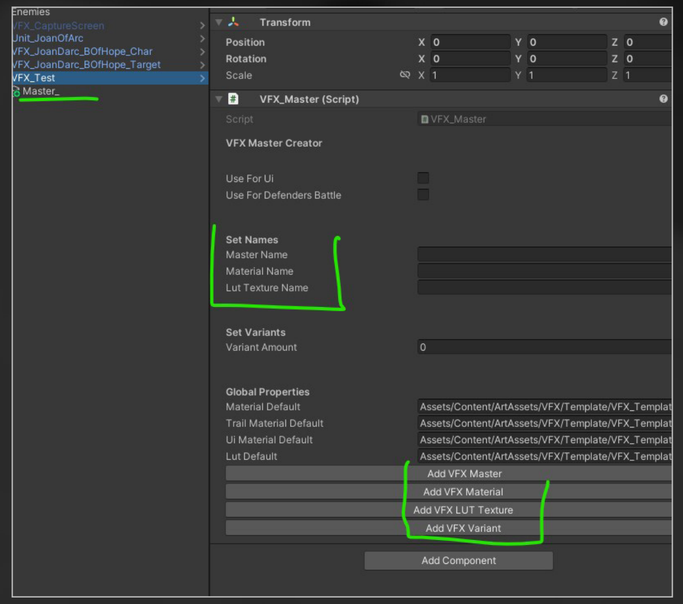
Add a VFX Object

1. To add a VFX Object to the project, right-click, go to VFX, and select ‘Create World Object.’
2. Give the VFX Object a name and place it in your hierarchy.
3. When you select your VFX Object, you will find a component on it called ‘VFX_Master.’ Click on ‘Add VFX Master,’ and your VFX Object will now become a Master object.

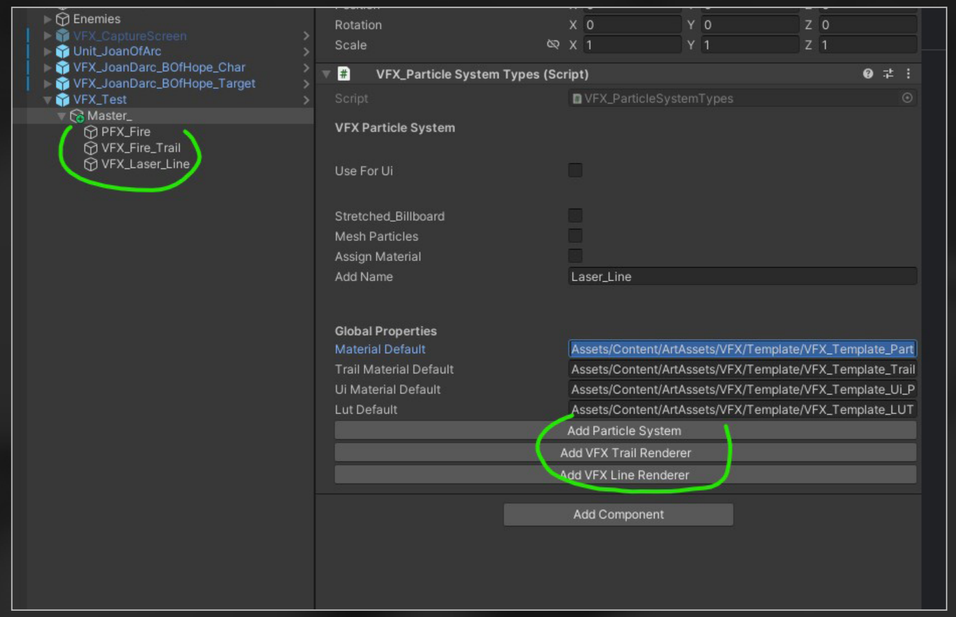
Here, you can add other assets to the project folder, such as Look Up Tables, Materials, and VFX Variants (which are prefab variants).







VFX Object Creation



- VFX Particle System Type

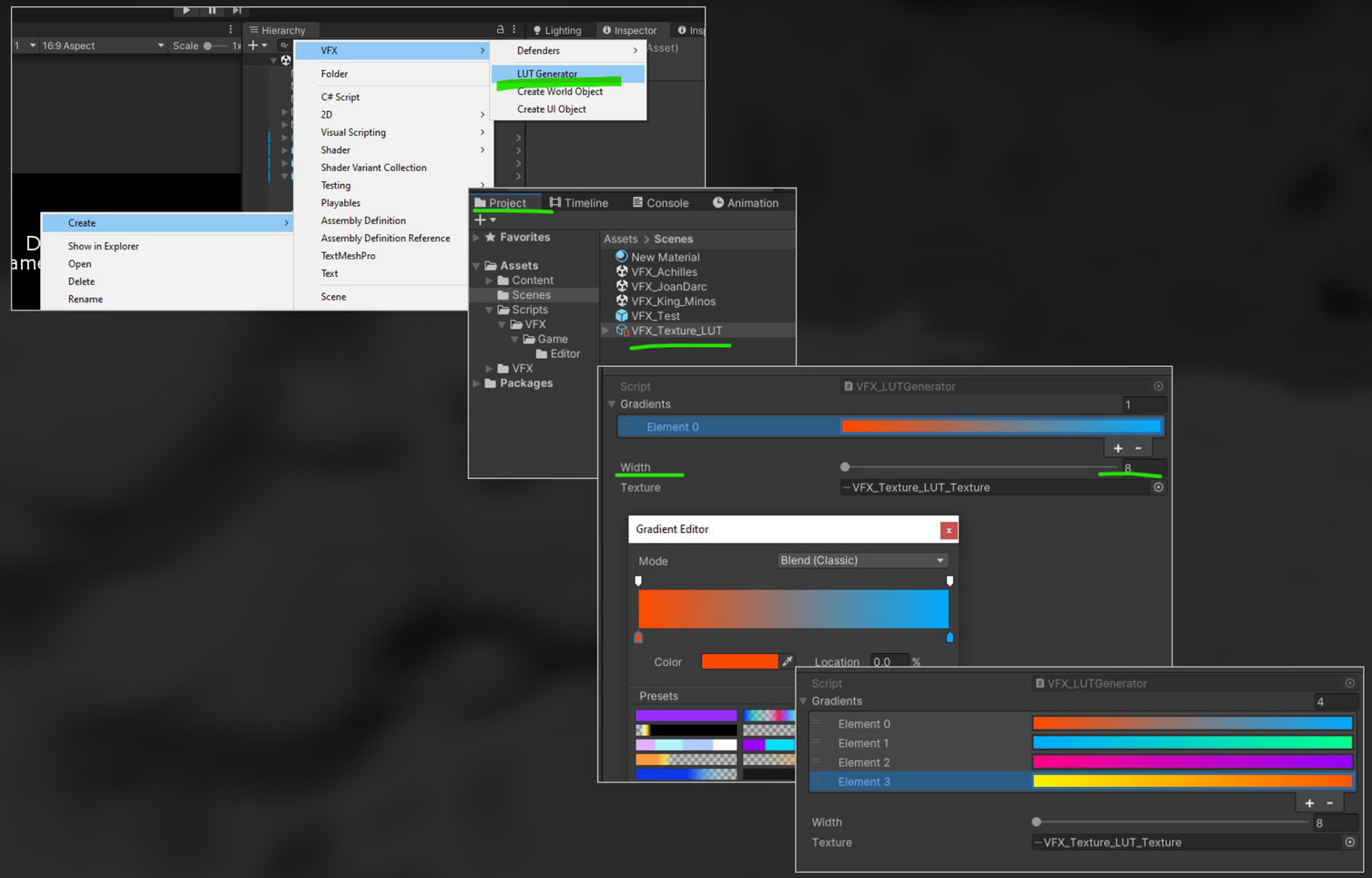
Your master has a component called ‘VFX Particle Systems Type.’ From here, you can add particle systems, Trail, and Line Renderer.

By enabling the ability to assign a material, your particle, trail, and line renderer will be associated with a material and a lookup texture.

If you need stretched or mesh particles, simply enable the toggle for that.

These particles will be set up with the correct vertex stream settings.

Look Up Textures Generator



- Add Look Up Texture

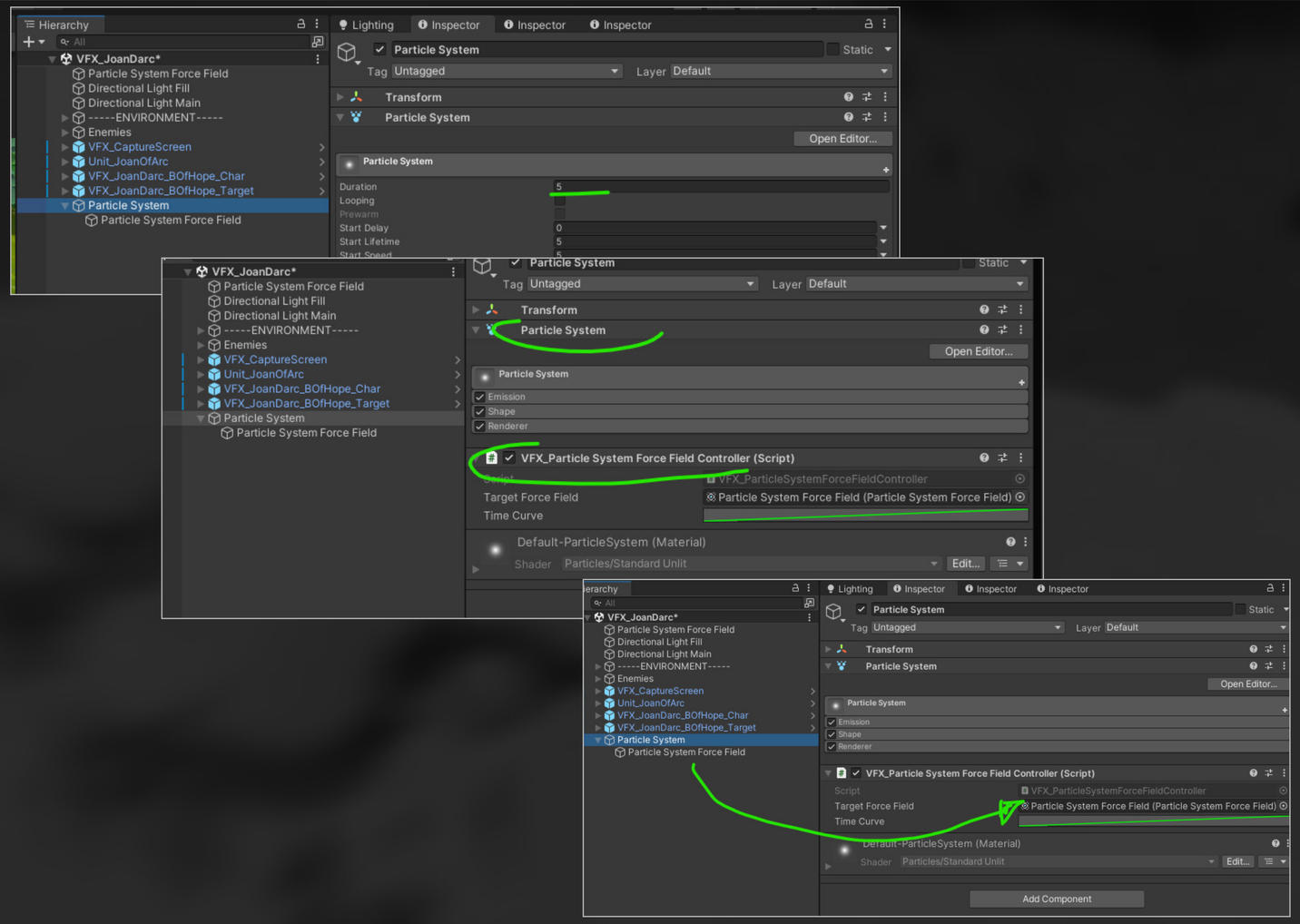
Right-click in the project to open the context menu.
Under VFX, locate 'LUT Generator.'

Click on it to add a new LUT asset. Upon selection, you will see a list of gradients.
Press the plus button to add a new one.

In certain shaders, you can animate the blending between the rows.

SCRIPTS

Particle System Force Field Controller

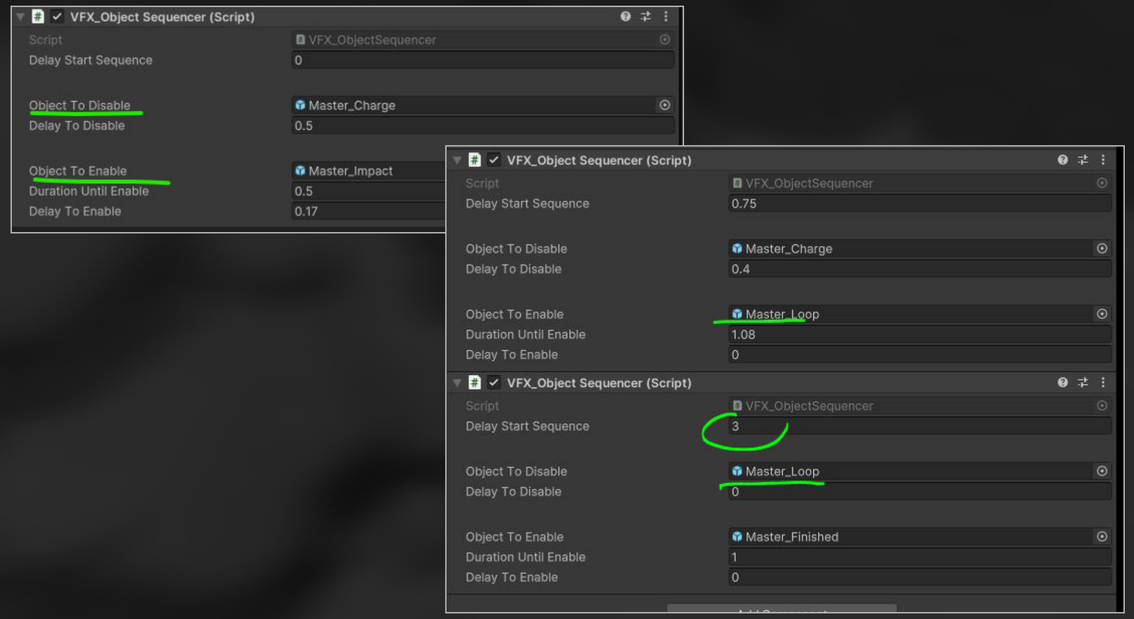


This script changes the gravity property of a Particle System Force Field over time using an AnimationCurve.

The intensity of the force field is controlled by the curve, providing a dynamic and animated effect to the associated Particle System.

Properties	Description
Target Force Field	**The target Particle System Force Field whose gravity property will be controlled by the script.**
Time Curve	**The AnimationCurve that defines the variation of the force field's gravity over time.**

Object Sequencer



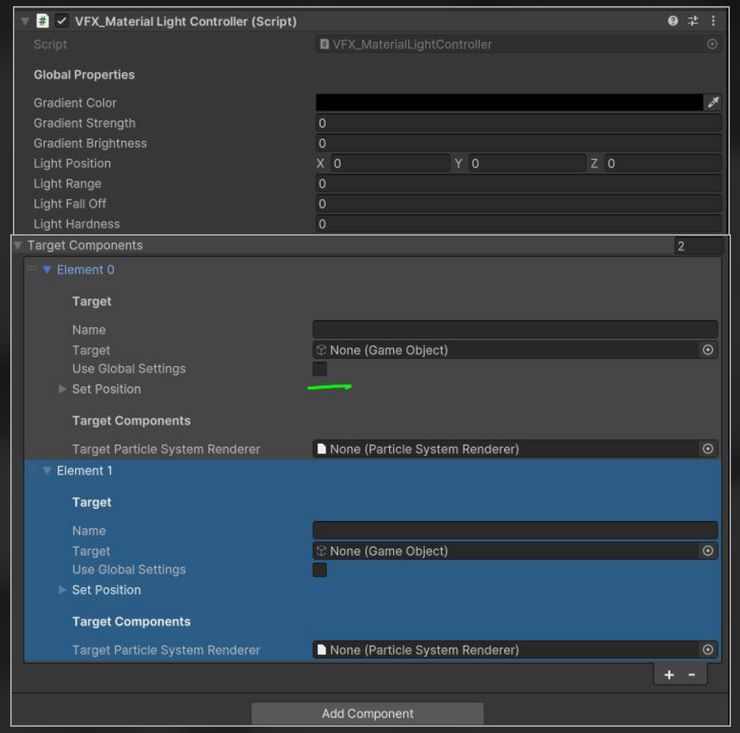
The script VFX Object Sequencer, is a component in Unity that facilitates the enabling and disabling of specified GameObjects over time.

The script ensures that designated GameObjects are activated and deactivated at specific intervals, creating a visual effects sequence.

NOTE: The object sequence should be on the root vfx object.

Properties	Description
Delay Start Sequence	The delay before the sequence starts.
Object To Disable	The GameObject to be disabled during the sequence.
Delay To Disable	The delay after the sequence starts before disabling the specified GameObject.
Object To Enable	The GameObject to be enabled during the sequence.
Duration Until Enable	The duration until the specified GameObject is enabled.
Delay To Enable	The delay after the sequence starts before enabling the specified GameObject.

Material Light Controller





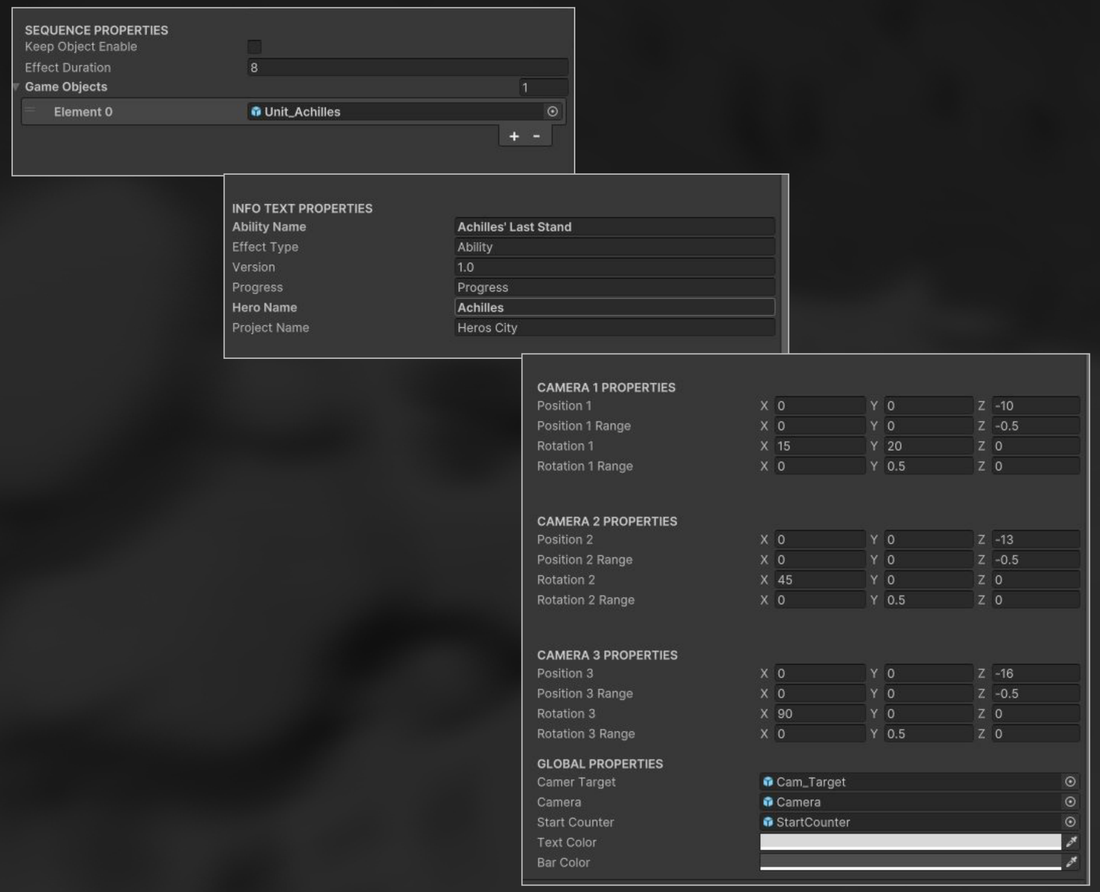
This script transfers the world position of the VFX object to the material property variable '_Set-Pos'.

If you are using Light Properties in your shader, you need to provide the shader with the world position data of your VFX object. Otherwise, your effect might appear incorrect.

Another feature is that this script can globally set the Light and Gradient Properties. This allows you to add one light source to many particle systems simultaneously.

Properties	Description
Global Properties	Check the page 'Gradient Properties'.
Target Components	
Name	Shows the name of the Particle System.
Target	Add here the Particle System.
Use Global Settings	For this element, the gradient and light properties will be overwritten by the global properties defined in the script.
Set Position	The updated world position from vfx object.
Target Particle System Renderer	The material from particle system.

Capture Screen

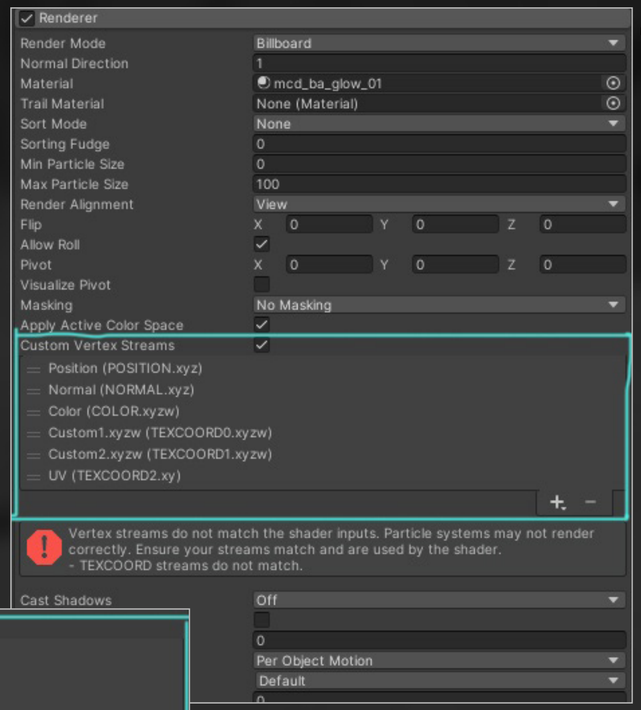


This script transfers the world position of the VFX object to the material property variable '_Set-Pos'.

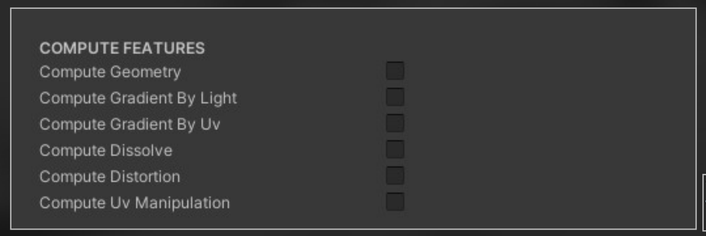
If you are using Light Properties in your shader, you need to provide the shader with the world position data of your VFX object. Otherwise, your effect might appear incorrect.

Another feature is that this script can globally set the Light and Gradient Properties. This allows you to add one light source to many particle systems simultaneously.

Properties	Description
Global Properties	Check the page 'Gradient Properties'.
Target Components	
Name	Shows the name of the Particle System.
Target	Add here the Particle System.
Use Global Settings	For this element, the gradient and light properties will be overwritten by the global properties defined in the script.
Set Position	The updated world position from vfx object.
Target Particle System Renderer	The material from particle system.



1.



3.



2.

VFX Shaders

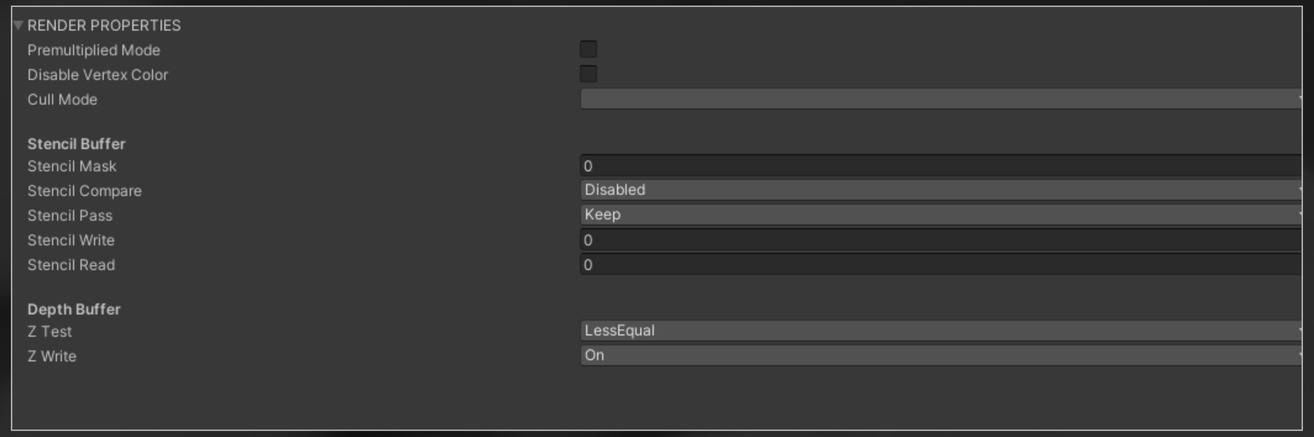
There are two main VFX shaders in the project: 'MultiColor Light And Motion' and an enhanced version for particles called 'Custom Data Light And Motion.'

Use the compute toggles to enable shader features such as dissolving, distortion, and UV manipulation, among others.

The custom data shader requires the Custom Data 1 module in the particle system. With custom data, you can manipulate specific properties over time for each particle.

NOTE: The vertex stream for billboards, stretched, vertical, and horizontal particles should look like in Picture 1.
For meshes, it should be like in the Picture 2.

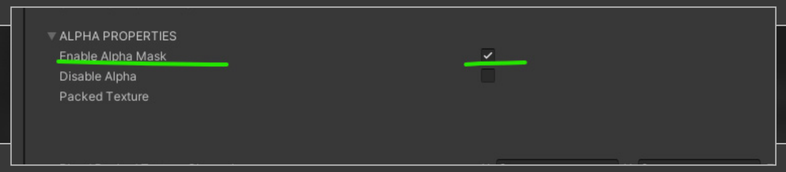
SHADER Render Properties



Properties	Description
Premultiplied Mode	Black areas in the alpha will blend additively.
Disable Vertex Alpha	Disabling the vertex alpha which is stored in the second UV's first index (X).
Cull Mode	Unity Documentation on Cull
Stencil Buffer	Configures the stencil buffer according to the given parameters.
Reference	Unity Stencil Documentation
Compare	Unity Stencil Documentation
Pass	Unity Stencil Documentation
Write	Unity Stencil Documentation
Read	Unity Stencil Documentation
Z Test	Unity ZTest Documentation
Z Write	Unity ZWrite Documentation

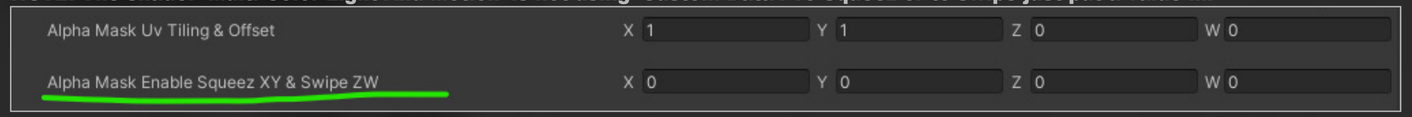
SHADER Alpha Properties







NOTE: The shader 'Multi Color Light And Motion' is not using 'Custom Data'. To Squeez or to swipe just put a value in.



Properties	Description
Alpha Properties	
Enable Alpha Mask	**Activates the alpha mask properties.**
Disable Alpha	**The packed texture will be extracted from alpha. This is necessary when you want to disconnect the grayscales from alpha and just to use it for the colors. The alpha mask can replace the alpha.**
Packed Texture	**Stores different masks in the channels, which are used as alpha.** **NOTE: Only the RGB channels are supported.**
Blend Packed Texture Channels	**Blends the different channels' XYZ values of RGB.**
Uv Tiling & Offset	**Uses UV tiling and offset to repeat and shift the texture's position.**
Alpha Mask	**Texture with grayscale information to define where and to what extent the visibility should be applied.**
use Alpha Uv	**Uv Tiling & Offset will used from packed textures.**
Uv Tiling & Offset	**Uses UV tiling and offset to repeat and shift the texture's position.**
Enable Squeez XY & Swipe ZW	**With a value of 1, it enables the use of a custom data value for squeezing and swiping the UV.**
Use Custom Data 1 Channel (0 - 3)	**Select a custom data channel from 0 to 3.**

SHADER Shading Properties



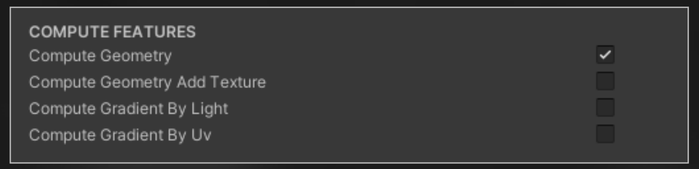
Properties	Description
Look Up Table	Map with a color gradient from left to right.
Switch LUT to Color Gradient	Gradient color will replaced by the LUT.
Multiply Particle Color	Particle / vertex colors will be added to mesh / particles.
Use custom data 1	Switch to control the v offset. For example to animate between two or more gradients.
CD1 Channel (0 - 3)	Select a custom data channel from 0 to 3.

SHADER World Position Mask Properties



Properties	Description
Enable World Position Mask	Multiplies the the world position gradient in to the alpha channel.
Set Look Up Table To World Position	Transfers the LUT colors in to the WP gradient.
Axis	Set direction of the WP mask.
Offset	Translate the world position.
Multiplier	Strength of the WP gradient.
Hardness	Sharpness of the WP gradient.

SHADER Geometry Properties







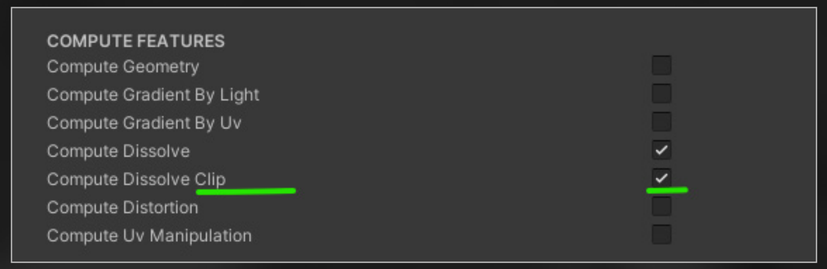
Properties	Description
Rim Multiplier	Strength of the rim effect.
Rim Hardness	Controls the hardness of the fall-off from the rim.
Rim Blend Strength	Controlls how mucht the rim is visible.
Deforming Frequency	Controls the frequency of the sinus curve.
Deforming Amplitude	Controls the strength of the sinus curve.
Deformation Axis	Translates the geometry to deform it.
Push Axis	Pushes the geometry along the normals.
Offset	Offsets the phase of the sinus curve.
Geo Offset Use Custom Data 1	Switch to control deformation strength by custom data 1.
Geo Offset By CD1 Channel (0 - 3)	Select a custom data channel from 0 to 3.

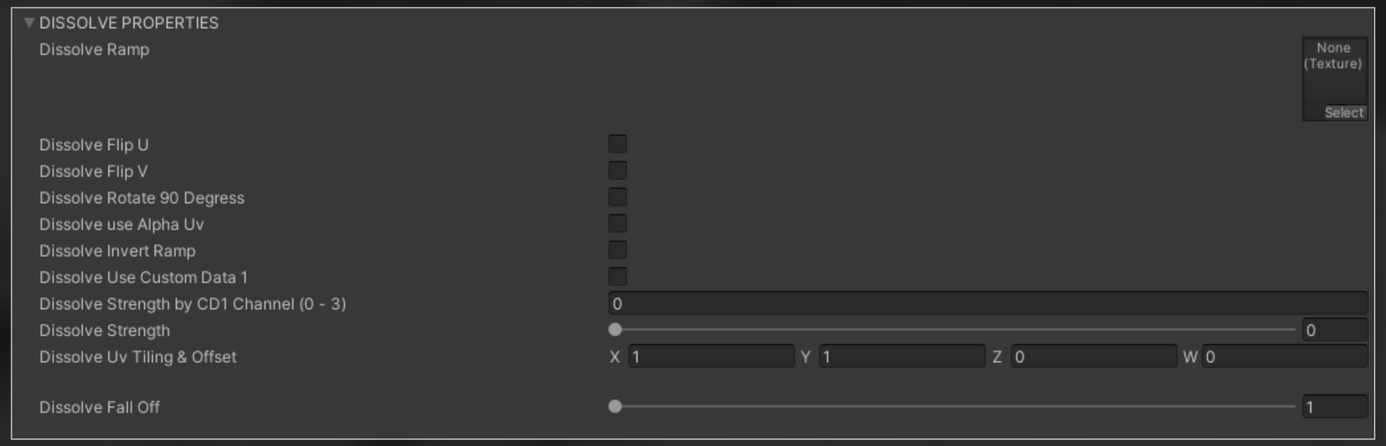
SHADER Gradient & Light Properties



Properties	Description
Color	**Sets the color of the gradient.**
Strength	**Controls the visibility of the gradient.**
Brightness	**Controls the strength of the adding effect.**
Blend Use Custom Data 1	**Switch to control the visibility of the gradient using custom data1.**
CD1 Channel (0 - 3)	**Select a custom data channel from 0 to 3.**
Gradient by Light	
Convert To A Light	**When this is enabled, the gradient adopts normal light behavior.**
Set Position	**Set the local poition.**
Light Position	**Set's the Light position relative to the local position.**
Light Range	**Controls the range of the light.**
Light Fall Off	**Higher values will create higer fall off range.**
Light Hardness	**Higher values will reduce the fall off.**
Gradient by Uv	
Uv Ramp Axis Blend	**Defines on wich direction the gradient should be create. X = Positive U, Y = Negative U, Z = Positive V, W = Negative V**
Uv Ramp Offset	**Translates the uv per axis.**

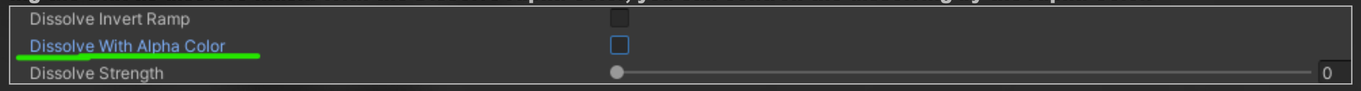
SHADER Dissolve Properties





Properties	Description
Map	Texture with grayscale information to define where and to what extent the dissolve should be applied.
Flip U	Flips the UV coordinates horizontally.
Flip V	Flips the UV coordinates vertically.
Rotate 90 Degrees	Rotates the UV coordinates by 90 degrees.
Use Alpha Uv	Uv Tiling & Offset will used from packed textures.
Invert Ramp	Inverts the gradients of the dissolve map.
Use Custom Data	With a value of 1, it enables the use of a custom data value for the dissolve strength.
Strength By Custom Data Channel (0 - 3)	Select a custom data channel from 0 to 3.
Strength	Controls the strength of the dissolve effect.
Uv Tiling & Offset	Uses UV tiling and offset to repeat and shift the texture's position.
Fall Off	Changes the values from the ramp; higher values will give hard edge results.

NOTE: The shader 'Multi Color Light And Motion' is not using 'Custom Data' to dissolve. In default state it is using the uv.x as dissolve mask. With the Dissolve Alpha Color, you can controll the dissolving by the Alpha Color.

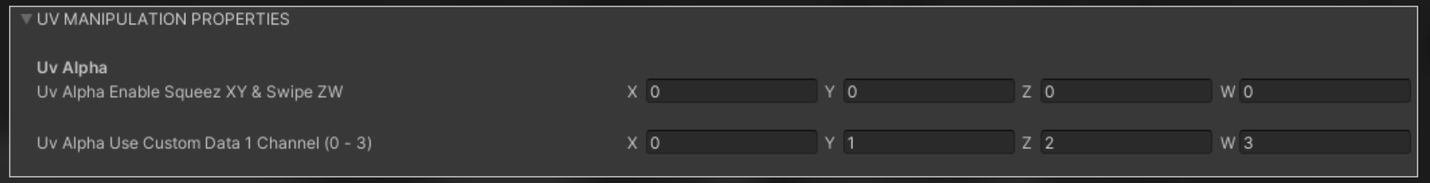


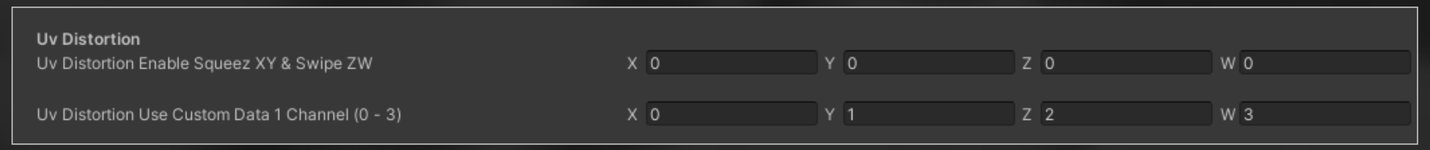
SHADER Distortion Properties



Properties	Description
Map	Texture with vector information applied to the UVs to create a directional distortion effect.
Mask	Texture with grayscale information to define where and to what extent the distortion should be applied.
Enable For Alpha	Distortion will be applied to the alpha texture.
Enable For Dissolve	Distortion will be applied to the dissolve map.
Use Custom Data 1	With a value of 1, it enables the use of a custom data value for the distortion strength.
Strength By Use Custom Data Channel (0 - 3)	Select a custom data channel from 0 to 3.
Strength	Controls the strength of the distortion effect.
Set Uv To World Coordinates	Set the uv's in to world coordinates.
Uv Tiling & Offset	Uses UV tiling and offset to repeat and shift the texture's position.

SHADER Uv Manipulation Properties



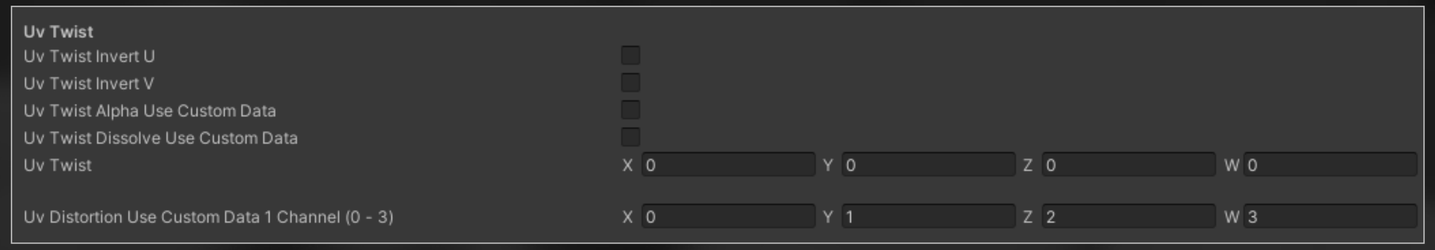


NOTE: The shader 'Multi Color Light And Motion' is not using 'Custom Data'. To Squeez or to swipe just put a value in.

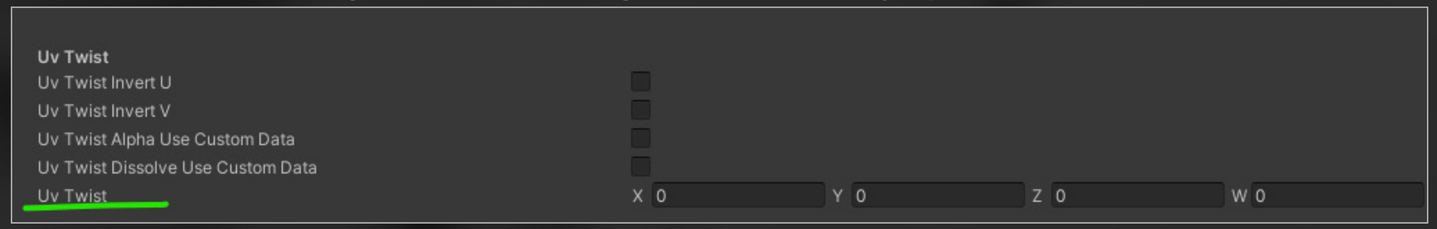


Properties	Description
Uv Alpha	
Enable Squeeze XY and Swipe ZW	**With a value of 1, it enables the use of a custom data value for squeezing and swiping the UV.**
Use Custom Data 1 Channel (0 - 3)	**Select a custom data channel from 0 to 3.**
Uv Distortion	
Enable Squeeze XY and Swipe ZW	**With a value of 1, it enables the use of a custom data value for squeezing and swiping the UV.**
Use Custom Data 1 Channel (0 - 3)	**Select a custom data channel from 0 to 3.**

SHADER Uv Manipulation Properties

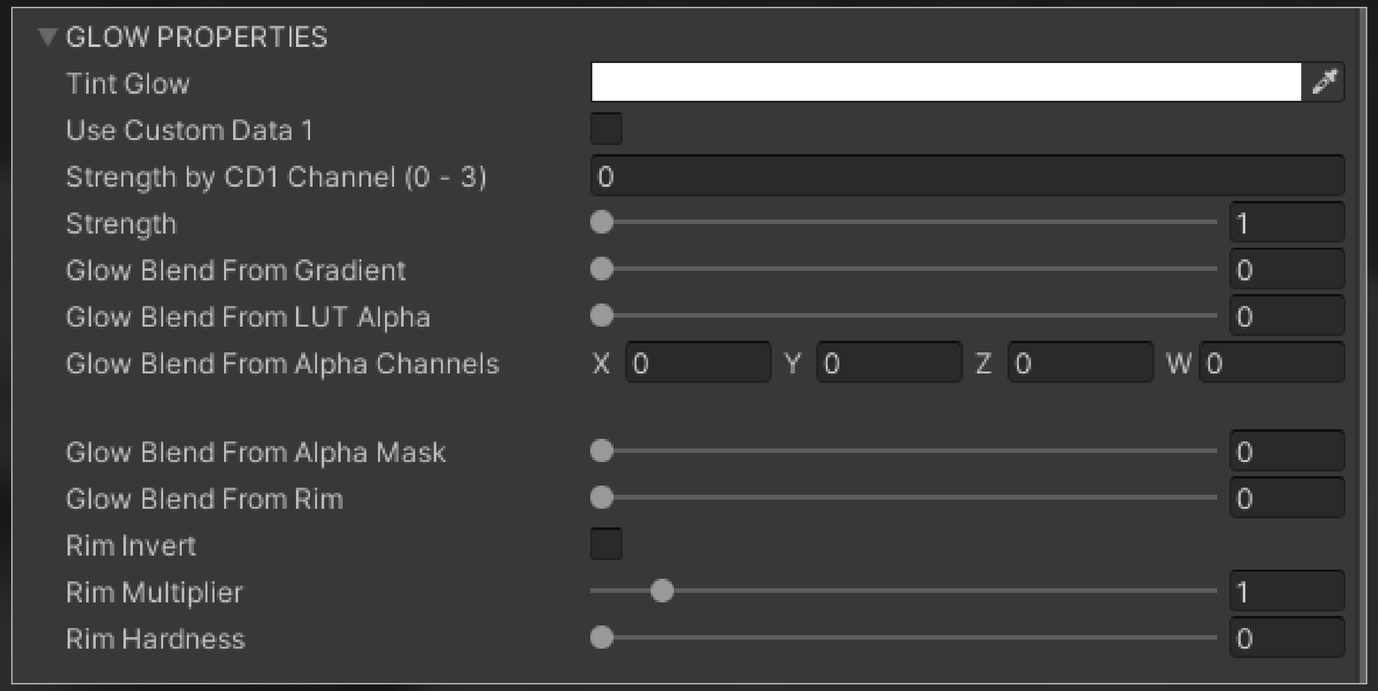


NOTE: The shader 'Multi Color Light And Motion' is not using 'Custom Data'. To twist just put a value in.



Properties	Description
Uv Twist	
Uv Twist Invert U	**Is inverting the u direction.**
Uv Twist Invert V	**Is inverting the v direction.**
Uv Twist Alpha Use Custom Data	**Adding the twist effect to the alpha texture.**
Uv Twist Dissolve Use Custom Data	**Adding the twist effect to the distortion map.**
Uv Twist	**Controlles the strength of the twist. If the Use Custom Data is Enabled. The four values becomes a switcher to activate the custom channel usage.**
Use Custom Data 1 Channel (0 - 3)	**Select a custom data channel from 0 to 3.**

SHADER Glow Properties



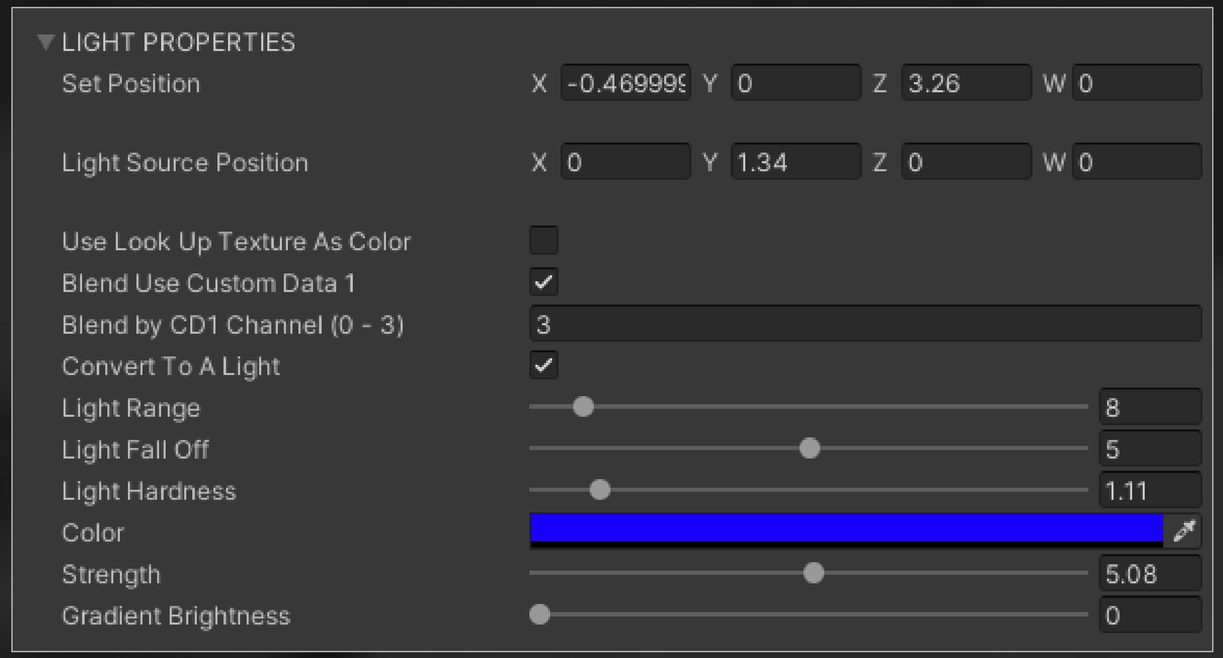
Properties	Description
Tint Glow	Sets the color of the glow.
Use Custom Data 1	Switch to control the visibility of the gradient using custom data1
Strength by CD1 Channel (0 - 3)	Select a custom data channel from 0 to 3.
Strength	Controls the strength of the glow.
Glow Blend From Gradient	Gradient features controls the visibility of the glow.
Glow Blend From LUT Alpha	Look Up Texture features controls the visibility of the glow.
Glow Blend From Alpha Channels	Alpha Channels features controls the visibility of the glow.
Glow Blend From Alpha Mask	Alpha Mask features controls the visibility of the glow.
Glow Blend From Rim	Rim features controls the visibility of the glow.
Rim Invert	Inverts the rim effect.
Rim Multiplier	Strength of the rim effect.
Rim Hardness	Controls the hardness of the fall-off from the rim.

Volume Shader



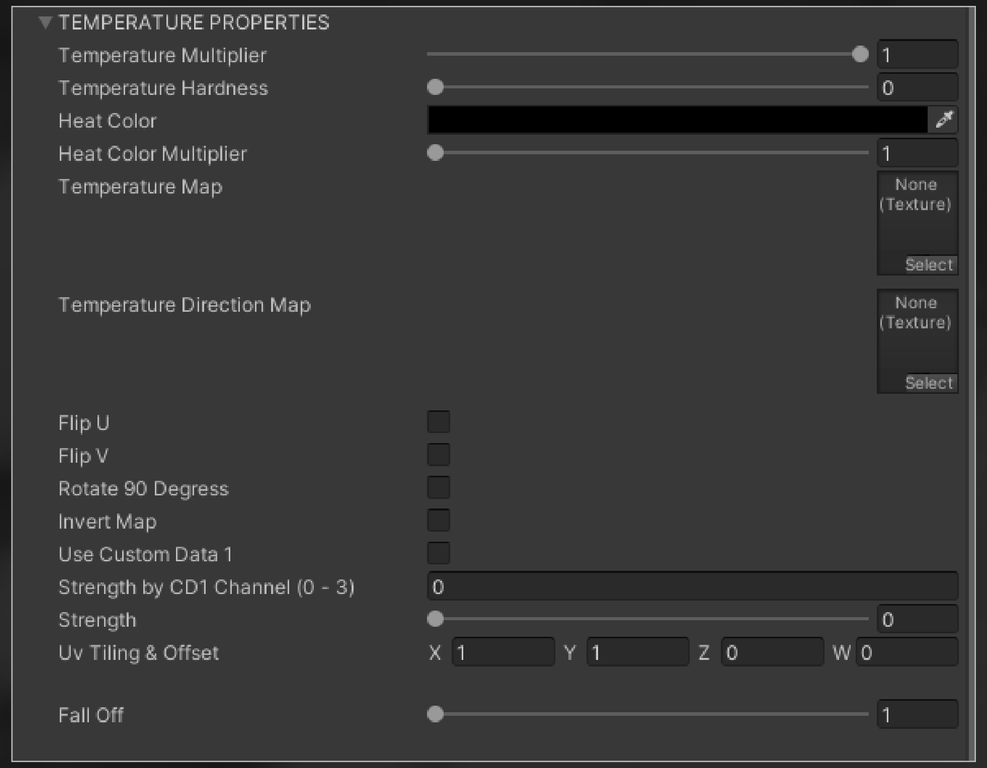
Properties	Description
Z Write Solid Layer	Layer will rendered in depth buffer.
Blend Solid Layer	Controls the visibility of the solid layer.
Squeez Solid Layer	Push the polys backwards along the normals.
Z Write Blur Layer	Layer will rendered in depth buffer.
Blend Blur Layer	Controls the visibility of the solid layer.
Rim Multiplier	Strength of the fresnel effect.
Rim Hardness	Higher values will create a sharper rim effect.
Look Up Table	Gradient to colorize the light.

Volume Shader



Properties	Description
Set Position	Set the local poition.
Light Soruce Position	Set the light position.
Use Look Up Texture As Color	Controls the strength of the adding effect.
Blend Use Custom Data 1	Switch to control the visibility of the gradient using custom data1.
Blend by CD1 Channel (0 - 3)	Select a custom data channel from 0 to 3.
Convert To Light	When this is enabled, the gradient adopts normal light behavior.
Light Range	Controls the range of the light.
Light Fall Off	Higher values will create higer fall off range.
Light Hardness	Higher values will reduce the fall off.
Color	Sets the light color.
Strength	Controls the visibility of the gradient.
Gradient Brigtness	Controls the strength of the adding effect.

SHADER Volume Shader



Properties	Description
Temperature Multiplier	Sets the range of Heat.
Tempperature Hardness	Sets the strength of the heat.
Heat Color	Is tinting the heat area.
Heat Color Multiplier	Overdrive the color values to generate glow.
Temperature Map	Grayscale texture to dissolve and to control the strength of the heat.
Temperature Direactio Map	Additional Grayscale texture to control the global dissolve direction.
Flip U	When this is enabled, the gradient adopts normal light behavior.
Flip V	Set the local poition.
Rotate 90 Degrees	Set's the Light position relative to the local position.
Invert Ramp	Controls the range of the light.
Use Custom Data	With a value of 1, it enables the use of a custom data value for the dissolve strength.
Strength By Custom Data Channel (0 - 3)	Select a custom data channel from 0 to 3.
Strength	Controls the strength of the dissolve effect.
Uv Tiling & Offset	Uses UV tiling and offset to repeat and shift the texture's position.
Fall Off	Changes the values from the ramp; higher values will give hard edge results.

SHADER Particle Instanced Mesh



Properties	Description
Compute Fog	Is adding the global fog.
MainTex	Color texture.
Tiling	UV tiling to repeat the texture's position..
Offset	UV offset to shift the texture's position.
Vertex Motion Settings	
Speed	Speed of the deformation.
Strength X	The strength of the deformation in the local X axis.
Use Symetry X	Mirrors the deformation of X.
Strength Y	The strength of the deformation in the local Y axis.
Use Symetry Y	Mirrors the deformation of Y.